

Network Forensics & triage

1. **The OSI Model Context:** In this lab, you analyzed the relationship between two specific layers of the Open Systems Interconnection (OSI) model:

Layer 7 (Application): DNS. This is where human-readable names are handled.

Layer 3 (Network): ICMP and IP. This is where routing, logical addressing, and connectivity diagnostics happen.

2. **DNS: The Internet's Phonebook (Layer 7):** Computers communicate using IP addresses, but humans use domain names (e.g., google.com). DNS translates these names.

The Transaction ID (The "Security Key"): DNS often uses UDP (User Datagram Protocol), which is fast but "connectionless." Because it doesn't establish a formal session, it is vulnerable to DNS Cache Poisoning (where a hacker sends a fake IP address to your laptop).

How it works: To prevent fakes, your laptop generates a random 16-bit Transaction ID (like the 0xe515 you found).

The Triage Rule: A legitimate response must contain the exact same Transaction ID as the query. If an analyst sees a response with a different ID, it is a massive red flag for a Man-in-the-Middle (MITM)

3. **ICMP: The Network's Diagnostic Tool (Layer 3):** ICMP is used by network devices to send error messages or operational information. The most common tool using ICMP is Ping.

TTL (Time To Live) & OS Fingerprinting: The TTL is a field in the IPv4 header that prevents packets from circulating forever in a loop. Every time a packet hits a router (a "hop"), the TTL is decremented by 1. If it hits 0, the packet is discarded.

Analysts use TTL for Passive OS Fingerprinting because different Operating Systems use different starting (Initial) TTL values:

Windows: Starts at 128.

Linux/Unix/Android: Starts at 64.

Network Infrastructure (Cisco): Often starts at 255.

Calculation: Since workstation is Windows, it sent a ping with a Start TTL of 128. Received a reply with a TTL of 113.

$128 \text{ (Initial)} - 113 \text{ (Observed)} = 15 \text{ Hops}$

This proves the destination was 15 routers away.

4. **VPN Encapsulation: The "Analyst Blind Spot"**

A VPN (Virtual Private Network) creates a secure "tunnel" to protect your privacy.

Encapsulation working:

Proton VPN takes your original DNS and ICMP packets and encapsulates (wraps) them inside a new, encrypted packet-usually using AES-256 encryption over UDP or TCP (SSL/TLS).

The Problem: To Wireshark, this looks like "Continuation Data" or "Encrypted Handshake." The analyst cannot see the internal headers (DNS names or Ping flags).

The Solution: Disabling the VPN "breaks" the tunnel, allowing the analyst to see the raw, unencapsulated traffic for forensic inspection.